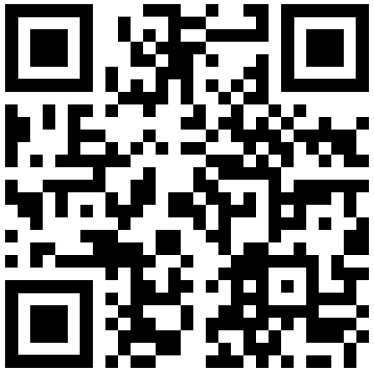
Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention

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https://arxiv.org/pdf/2006.16236



https://linear-transformers.com/

Why Linear Attention?



$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$$

$$Q, K, V \in \mathbb{R}^{N \times d}$$

$$O(N^2 d)$$

One does not simply…

$$(QK^T)V = Q(K^TV)$$

$$\text{softmax}(QK^T)V \neq Q\,\text{softmax}(K^TV)$$





Generalized Attention

$$y_i = \frac{\sum_{j=1}^{N} \exp\left(\frac{q_i^T k_j}{\sqrt{d}}\right) v_j}{\sum_{j=1}^{N} \exp\left(\frac{q_i^T k_j}{\sqrt{d}}\right)} \longrightarrow y_i = \frac{\sum_{j=1}^{N} \mathrm{sim}(q_i, k_j) v_j}{\sum_{j=1}^{N} \mathrm{sim}(q_i, k_j)}$$

$$\mathrm{sim}(q, k) \geq 0$$

Linearized Attention

$$\text{sim}(q, k) = \phi(q)^T \phi(k) \qquad \phi : \mathbb{R}^d \to \mathbb{R}^d$$

$$y_i = \frac{\sum_{j=1}^{N} \phi(q_i)^T \phi(k_j) v_j}{\sum_{j=1}^{N} \phi(q_i)^T \phi(k_j)} \quad \Rightarrow \quad y_i = \frac{\phi(q_i)^T \sum_{j=1}^{N} \phi(k_j) v_j^T}{\phi(q_i)^T \sum_{j=1}^{N} \phi(k_j)}$$

$$\left(\phi(Q)\phi(K)^T\right)V = \phi(Q)\left(\phi(K)^T V\right)$$

Linearized Attention

$$y_i = \frac{\phi(q_i)^T \sum_{j=1}^{N} \phi(k_j) v_j^T}{\phi(q_i)^T \sum_{j=1}^{N} \phi(k_j)}$$

$$S = \sum_{j=1}^{N} \phi(k_j) v_j^T \qquad z = \sum_{j=1}^{N} \phi(k_j)$$

$$y_i = \frac{\phi(q_i)^T S}{\phi(q_i)^T z}$$

Computational Cost

$$\phi(Q), \phi(K), V \in \mathbb{R}^{N \times d}$$

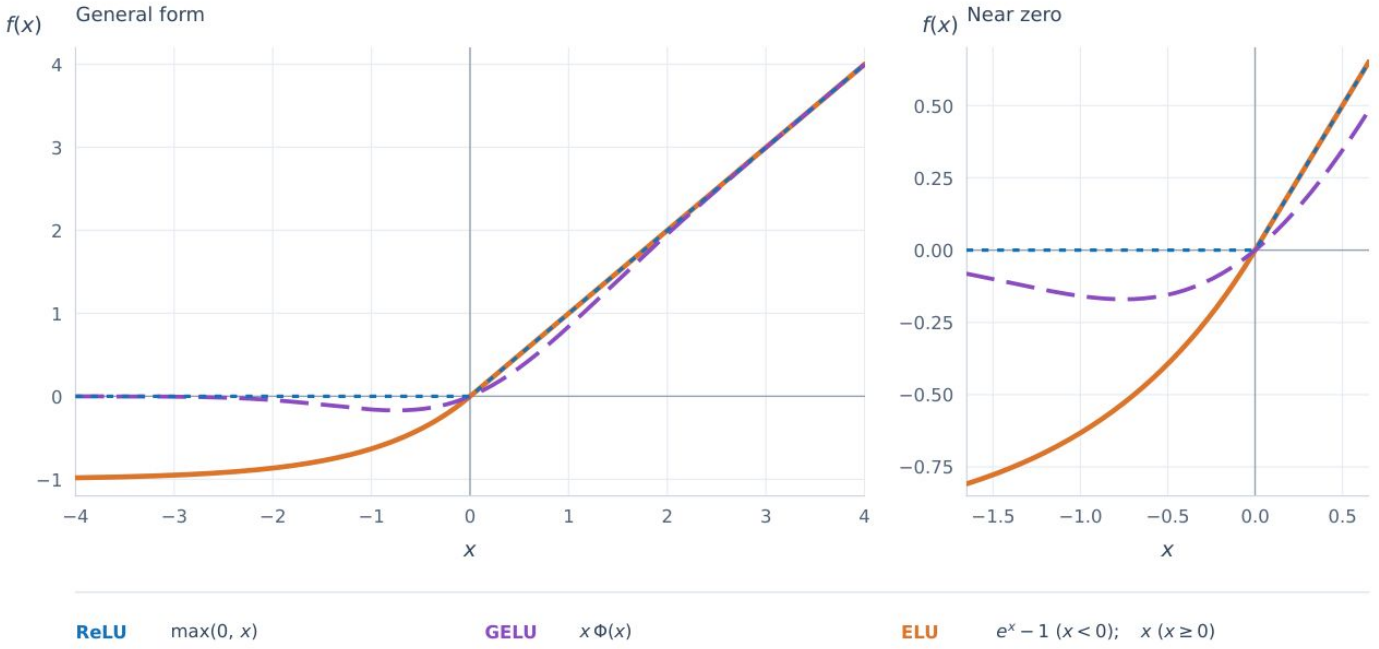
$$\underbrace{\left(\phi(Q)\phi(K)^T\right)}_{N \times N} V = \phi(Q) \underbrace{\left(\phi(K)^T V\right)}_{d \times d}$$



ELU+1

$$\phi(x) = \mathrm{ELU}(x) + 1$$

$$\phi(x) = \begin{cases} x + 1 & x \geq 0 \\ e^x & x < 0 \end{cases}$$



Causal Linear Attention

$$y_i = \frac{\phi(q_i)^T \sum_{j=1}^{i} \phi(k_j) v_j^T}{\phi(q_i)^T \sum_{j=1}^{i} \phi(k_j)}$$

$$S_i = \sum_{j=1}^{i} \phi(k_j) v_j^T \quad z_i = \sum_{j=1}^{i} \phi(k_j)$$

$$y_i = \frac{\phi(q_i)^T S_i}{\phi(q_i)^T z_i}$$

Transformers are RNNs

$$S_0 = 0$$

$$z_0 = 0$$

$$S_i = S_{i-1} + \phi(k_i)v_i^T$$

$$z_i = z_{i-1} + \phi(k_i)$$

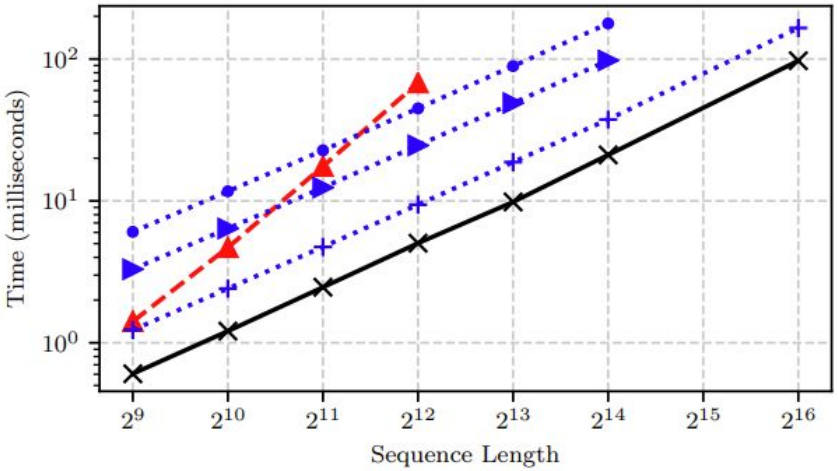
$$y_i = \frac{\phi(q_i)^T S_i}{\phi(q_i)^T z_i}$$

- The state (S_i, z_i) aggregates information from all previous tokens
- Processing a new token requires only the previous state and the current q_i, k_i, v_i
- The state has a fixed size of O(d^2) per attention head, independent of sequence length

$S_i \in \mathbb{R}^{d \times d}$ $z_i \in \mathbb{R}^d$

A Causal Linear Transformer is equivalent to an RNN with a fixed-size hidden state

Computational Efficiency



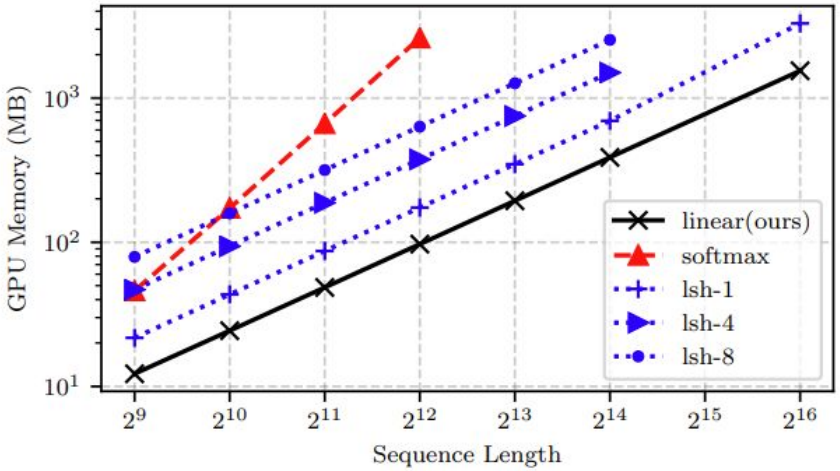


Figure 1: Comparison of the computational requirements for a forward/backward pass for Reformer (lsh-X), softmax attention and linear attention. Linear and Reformer models scale linearly with the sequence length unlike softmax which scales with the square of the sequence length both in memory and time. Full details of the experiment can be found in § 4.1.

Autoregressive Image Generation

Model	Images/sec ↑	Bits/dim ↓
Softmax	0.004	3.47
Softmax + KV-cache	0.32	3.47
Linear Attention	**17.85**	**3.40**

- Higher throughput than the standard Softmax baseline
- Comparable generation quality with significantly lower computational cost

The ladder: five years, one line of math

Year	Model	Update rule for the state S	What it adds
2020	Linear Attention	$S_t = S_{t-1} + v_t k_t^\top$	$O(n)$, but never forgets
2023	RetNet	$S_t = \gamma\, S_{t-1} + v_t k_t^\top$	fixed exponential decay
2024	GLA, Mamba-2	$S_t = S_{t-1}\,\mathrm{Diag}(\alpha_t) + v_t k_t^\top$	a *learned* gate
2024	DeltaNet	$S_t = S_{t-1}\,(I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top$	error-correcting write
2025	Gated DeltaNet	$S_t = \alpha_t\, S_{t-1}\,(I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top$	gate *and* delta
2025	**KDA** (Kimi)	$S_t = S_{t-1}\,\mathrm{Diag}(a_t)\,(I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top$	**channel-wise** gate

All of 2025's "magic" is the evolution of a single line. Each row adds exactly one term.

RetNet [27] · GLA [35] · Mamba-2 [10] · DeltaNet [36] · Gated DeltaNet [34] · KDA [21]

Sources

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